

One-to-One table matching between the revised Argentinean data subset and the consolidated database.

Cristian Correa and Maryse Boisjoly

2026-05-30

Contents

1	Introduction	1
2	Load Required Libraries	2
3	Data importation	2
3.1	Current database	2
3.2	Revised database received back from Daniela Milano and Juan Barriga	2
4	Preliminary handling	2
5	Diagnostic check of matching fields	3
6	Making the match based on exact catalogNumber match	4
7	Manual revision of ambiguous matches	4
8	Final table matching	5
9	Session Information	6

1 Introduction

Aquí se documenta cómo se vincularon los comentarios y revisiones de Daniela Milano y Juan Pablo Barriga sobre las ocurrencias reportadas en territorio Argentino.

Este documento describe el método empleado para restablecer el vínculo uno-a-uno entre la tabla de comentarios y revisiones (table1) y la de la base de datos (table2) de modo de poder ejecutar dichos cambios, asegurando correspondencias precisas y completamente trazables.

Principios clave del flujo de trabajo:

- Cada fila de `table1` debe coincidir de manera exacta con una única fila de `table2`.
- Las comparaciones se realizan exhaustivamente contra todas las filas de la otra tabla.

La correspondencia se efectuó usando el campo `catalogNumber`, salvo por 12 filas que se revisaron y vincularon manualmente.

2 Load Required Libraries

```
setwd("C:/Users/crist/Documents/GitHub/Atlas_Ictiogeografico/table_joining")

rm(list=ls())

library(dplyr)
library(stringr)
library(ggplot2)
library(knitr)
library(tidyr)
library(kableExtra)
```

3 Data importation

3.1 Current database

```
# Load data to start from here
```

```
X.filtered.dedup <- read.csv("../outputs_global_analyses/dataset_filtered_deduplicated_after_manual_correc
```

3.2 Revised database received back from Daniela Milano and Juan Barriga

```
CM.Ar <- read.csv("../argentina/AtlasIctio2025_Datos_DwC_deInteres_norepetidos DM_Sfontinalis.csv",
  na.strings = c("", "NA"))
```

4 Preliminary handling

```
## Fix character encoding-related issues.
```

```
CM.Ar <- CM.Ar %>%
  mutate(across(
    .cols = where(is.character) &
      !any_of("correccion") &
      !starts_with("."),
    .fns = ~ iconv(., from = "UTF-8", to = "Windows-1252", sub = "byte")
  ))
```

```
# Merge (coalesce) two columns that were separated only due to a typo.
```

```
CM.Ar <- CM.Ar %>%
  mutate(
    across(c(bibliographicCitation, bibliographicCitation),
      ~ str_trim(.) %>% na_if("") %>% na_if("NA")),
    bibliographicCitation = coalesce(bibliographicCitation, bibliographicCitation)
  ) %>% select(-bibliographicCitation)
```

```
# Remove "NUEVOS" because these enter through another route, and sort by id_CC
CM.Ar <- CM.Ar %>% filter(id_CC != "NUEVO") %>%
  arrange(as.numeric(id_CC))
```

Exclude rows from CM.Ar that do not have changes to apply (no action required)

```
# Keep only rows with comments in another object
CM.Ar <- CM.Ar %>%
  filter(!is.na(correccion))
```

Exclude data from Fundación Óga (institutionCode == "FO"), because these were imported from another, better source.

```
CM.Ar <- CM.Ar %>% filter(institutionCode != "FO" | is.na(institutionCode))

# %>% select(datasetName)
```

5 Diagnostic check of matching fields

Select only the candidate matching field from both tables. Add row IDs so that each row can be tracked unambiguously during checks.

```
CM.Ar.match <- CM.Ar %>%
  mutate(CM_row_id = row_number()) %>%
  select(CM_row_id, catalogNumber, everything())

X.master.match <- X.filtered.dedup %>%
  mutate(master_row_id = row_number())
```

Count how many times each catalogNumber appears in CM.Ar.

```
CM.catalog_counts <- CM.Ar.match %>%
  count(catalogNumber, name = "n_CM")

# Count how many times each catalogNumber appears in the master table.

master.catalog_counts <- X.master.match %>%
  count(catalogNumber, name = "n_master")

# Combine the counts to identify clean, ambiguous, and unmatched cases.

catalog_match_check <- CM.Ar.match %>%
  select(CM_row_id, catalogNumber) %>%
  left_join(CM.catalog_counts, by = "catalogNumber") %>%
  left_join(master.catalog_counts, by = "catalogNumber") %>%
  mutate(
    n_master = replace_na(n_master, 0),
    match_status = case_when(
      is.na(catalogNumber) ~ "missing catalogNumber",
      n_CM == 1 & n_master == 1 ~ "one-to-one",
      n_CM > 1 & n_master >= 1 ~ "duplicated in CM.Ar",
      n_CM == 1 & n_master > 1 ~ "multiple matches in master",
      n_master == 0 ~ "no match in master",
```

```

    TRUE ~ "check manually"
  )
)

```

Summarize how many rows fall into each matching category.

```

catalog_match_check %>%
  count(match_status)

```

```

##           match_status    n
## 1  missing catalogNumber    2
## 2 multiple matches in master    1
## 3      no match in master    9
## 4           one-to-one 318

```

6 Making the match based on exact catalogNumber match

Keep only the cases where catalogNumber gives exactly one match in CM.Ar and exactly one match in the master database.

```

CM.Ar.clean_matches <- CM.Ar.match %>%
  left_join(
    catalog_match_check %>%
      select(CM_row_id, match_status),
    by = "CM_row_id"
  ) %>%
  filter(match_status == "one-to-one") %>%
  left_join(
    X.master.match %>%
      select(prim_key,
             master_row_id,
             catalogNumber,
             occurrenceID_master = occurrenceID,
             canonicalName_master = canonicalName,
             locality_master = locality,
             datasetName_master = datasetName,
             eventDate_master = eventDate,
             decimalLatitude_master = decimalLatitude),
    by = "catalogNumber")

```

Note: Several sanity checks (not shown) demonstrated adequate matching between the two tables.

7 Manual revision of ambiguous matches

The next step was to extract 12 cases that could not be matched automatically, revise them manually (not shown), and crosscheck with the master database to establish the links by indexing the `prim_key`. If no valid match is found, leave `prim_key` blank.

These should be reviewed manually.

```

CM.Ar.manual_check <- CM.Ar.match %>%
  left_join(
    catalog_match_check %>%
      select(CM_row_id, match_status),
    by = "CM_row_id"
  ) %>%
  filter(match_status != "one-to-one")

manual_link <- data.frame(CM_row_id = c(5,11,13,81,83,200,263,274,301,306,321,324),
  prim_key = c("agui | MAguilar.Lit.Review.1424",
    "gbif | urn:catalog:CAS:ICH:63844",
    "gbif | FFPAT086-11",
    "inat | https://www.inaturalist.org/observations/147830684",
    "gbif | KJ499172",
    NA,
    "gbif | KJ499176",
    "gbif | FFPAT042-11",
    NA,
    NA,
    NA,
    NA))

CM.Ar.manual_check <- CM.Ar.manual_check %>%
  left_join(manual_link, by = "CM_row_id") %>% relocate(prim_key)

```

8 Final table matching

Finally, combine both the automatic assignments and manual assignments, and consolidate the table of corrections to be applied to the master database in the main code.

```

CM.Ar.link <- CM.Ar.clean_matches %>% bind_rows(CM.Ar.manual_check) %>% select(CM_row_id, prim_key)

Cambios_a_implementar_Argentina <- CM.Ar %>% mutate(CM_row_id = row_number()) %>%
  select(CM_row_id, correccion, RemoveFromMaster, starts_with(".")) %>%
  left_join(CM.Ar.link, by = "CM_row_id") %>% relocate(prim_key, .after = CM_row_id)

```

Table 1: Sample of the finished table of corrections. The second column corresponds to the primary key of the master database.

CM_row_id	prim_key	correccion	RemoveFromMaster	Location	.decimalLongitude
1	gbif APN-PM-849	Locality: Lago Azara	NA	Lago Azara	NA
2	agui MAguilar.Lit.Review.1163	coord 48.9978/72.2456	NA	NA	-48.99780
3	agui MAguilar.Lit.Review.1182	coord correctas 49.0093/72.2437	NA	NA	-49.00930
4	agui MAguilar.Lit.Review.471	locality Lagura Pescado; coord 147.82058/72.1659	NA	Lagura Pescado	-47.82058
5	agui MAguilar.Lit.Review.1424	coord 47.82/72.118; year 2001; Recorded By D Milano; individual Count 132	NA	NA	-47.82000

CM_row	id_key	correccion	RemoveFromMass	id	.decimalLongitude
6	agui MAguilar.Lit.Review.951	coord 47.82/72.118; year 2001; RecordedBy D Milano; individualCount132	NA	NA	-47.82000

Save correction table to hard drive

```
write.csv(Cambios_a_implementar_Argentina, "Cambios_a_implementar_Argentina_2026-05-30.csv", fileEncoding="utf8")
```

9 Session Information

```
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_Canada.utf8  LC_CTYPE=English_Canada.utf8
## [3] LC_MONETARY=English_Canada.utf8 LC_NUMERIC=C
## [5] LC_TIME=English_Canada.utf8
##
## time zone: America/Santiago
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.0 tidyr_1.3.1      knitr_1.50      ggplot2_4.0.1
## [5] stringr_1.6.0    dplyr_1.1.4
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      compiler_4.4.1    tidyselect_1.2.1  tinytex_0.58
## [5] xml2_1.5.0        systemfonts_1.3.1 scales_1.4.0      textshaping_1.0.4
## [9] yaml_2.3.10       fastmap_1.2.0     R6_2.6.1          generics_0.1.4
## [13] tibble_3.3.0      svglite_2.2.2     pillar_1.11.1     RColorBrewer_1.1-3
## [17] rlang_1.1.6       stringi_1.8.7     xfun_0.54         S7_0.2.1
## [21] viridisLite_0.4.2 cli_3.6.5         withr_3.0.2       magrittr_2.0.4
## [25] digest_0.6.39     grid_4.4.1        rstudioapi_0.17.1 lifecycle_1.0.4
## [29] vctrs_0.6.5       evaluate_1.0.5    glue_1.8.0        farver_2.1.2
## [33] rmarkdown_2.30    purrr_1.2.0       tools_4.4.1       pkgconfig_2.0.3
## [37] htmltools_0.5.8.1
```